Python-BeautifulSoup

Objective

- Introduction
- Steps involved in web scraping
- third-party libraries
- web scraping Process
- Summary

Introduction

Access the HTML of the webpage and extract useful information/data from it. This technique is called web scraping or web harvesting or web data extraction.

There are mainly two ways to extract data from a website:

1. Use the API of the website (if it exists). For example, Facebook has the Facebook Graph API which allows retrieval of data posted on Facebook.
2. Implementation of a Web Scraping framework of Python called **Beautiful Soup.**

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Steps involved in web scraping

1. Send an HTTP request to the URL of the webpage.
2. The server responds to the request by returning the HTML content of the webpage. Third-party HTTP library used for python-requests .
3. After accessing the HTML content, need to parsing the data. Since most of the HTML data is nested we have to create a nested/tree structure of the HTML data. There are many HTML parser libraries available but the most advanced one is html5lib.
4. Now, navigate and search the parse tree i.e. tree traversal. For this task, we will be using Beautiful Soup. It is a Python library for pulling data out of HTML and XML files.

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Third-party libraries

- We have to install following libraries :
  - pip install requests
  - pip install html5lib
  - pip install beautifulsoup4

If you are using Anaconda

- Start Navigator.
- In Environments tab checked for installed packeges.

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Web scraping Process

- **Accessing the HTML content from webpage**

```
import requests
URL = "https://www.know-it.co.in//"
r = requests.get(URL)
print(r.content)
```

- Parsing the HTML content

```
soup = BeautifulSoup(r.content, 'html5lib') # If this line causes an
error, run 'pip install html5lib' or install html5lib
print(soup.prettify())
```


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- **Searching and navigating through the parse tree**

```
import requests
from bs4 import BeautifulSoup
url="http://www.values.com/inspirational-quotes"
r= requests.get(url)
soup = BeautifulSoup(r.content,'html5lib')

quotes=[]
tbl= soup.find('div',attrs={'id':'all_quotes'})

for row in tbl.findAll('div',attrs={'class':'col-6 col-lg-3 text-center margin-30px-
bottom sm-margin-30px-top'}):
    qu={}
    qu['text']=row.img['alt'].split("#")[0]
    qu['Author']=row.img['alt'].split("#")[1]
    quotes.append(qu)

cnt=0
for q in quotes:
    cnt+=1
    print(cnt,q['text'],end=' ')
```

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Thank You